

2. ER Diagram & Design

Quick Commerce System

2.1 Identification of Entities

Based on the case study and system scope, the following **core entities** are identified:

1. **Customer**
 2. **Product**
 3. **Order**
 4. **Order_Item** (Associative / Weak Entity)
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2.2 Entity Description with Attributes

1. Customer

Represents users who place orders on the quick commerce platform.

Attributes:

- **customer_id** (Primary Key)
 - customer_name
 - phone_number
 - email
 - address
 - registration_date
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2. Product

Represents items available for instant delivery.

Attributes:

- **product_id** (Primary Key)
- product_name
- category
- price
- stock_quantity

- availability_status
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3. Order

Represents a purchase transaction made by a customer.

Attributes:

- **order_id** (Primary Key)
 - order_date
 - order_status
 - total_amount
 - **customer_id** (Foreign Key)
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4. Order_Item

Represents products included in a specific order
(Resolves the many-to-many relationship between Order and Product).

Attributes:

- **order_item_id** (Primary Key)
 - quantity
 - price_at_order
 - **order_id** (Foreign Key)
 - **product_id** (Foreign Key)
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2.3 Relationships & Cardinalities

1. Customer — Places — Order

- **Relationship:** Customer *places* Order
 - **Cardinality:**
 - One Customer → Many Orders
 - Each Order → Exactly One Customer
 - **Type:** One-to-Many (1:N)
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2. Order — Contains — Product

- **Relationship:** Order *contains* Product
 - **Cardinality:**
 - One Order → Many Products
 - One Product → Can appear in Many Orders
 - **Type:** Many-to-Many (M:N)
 - **Resolved Using:** Order_Item entity
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3. Order — Has — Order_Item

- **Cardinality:**
 - One Order → Many Order_Items
 - Each Order_Item → One Order
 - **Type:** One-to-Many (1:N)
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4. Product — Appears In — Order_Item

- **Cardinality:**
 - One Product → Many Order_Items
 - Each Order_Item → One Product
 - **Type:** One-to-Many (1:N)
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2.4 ER Diagram Notation (Exam-Oriented)

- **Rectangles** → Entities (Customer, Product, Order, Order_Item)
 - **Ellipses** → Attributes
 - **Underlined attributes** → Primary Keys
 - **Diamonds** → Relationships (Places, Contains, Has)
 - **Double rectangle** → Weak/Associative entity (Order_Item)
 - **Crow's Foot / 1:N notation** → Cardinalities
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2.5 Design Justification

- The ER model is **conceptually correct and complete**
- All **entities and attributes are logically identified**
- **Primary and foreign keys** ensure referential integrity
- **Many-to-many relationship** is properly resolved

- Design supports **SQL*Plus table creation without redundancy**
- Easily extendable for delivery, payments, or reviews